

CoRoT-1b

R-2550

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_190329 · created 2026-08-26T00:51:21+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458571.6732 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.027
Transit depth (Rp/Rs)^2	2.03 +/- 0.78 [%]
Orbital Inclination (inc)	80.55 +/- 2.41
Airmass coefficient 1 (a1)	1.41 +/- 0.045
Airmass coefficient 2 (a2)	-0.1413 +/- 0.0067
Scatter in the residuals of the lightcurve fit is	3.09 %
Best Comparison Star	None
Optimal Aperture	3.42
Optimal Annulus	16.22
Transit Duration (day)	0.078 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.027 vs 0.1388 (deviation 0.09σ)
Duration fit vs published	0.078 d vs 0.1046 d (deviation 0.83σ)
Mid-time O–C	-5.9 min
Depth SNR	3.9
Residual scatter	3.09 %

Light curve

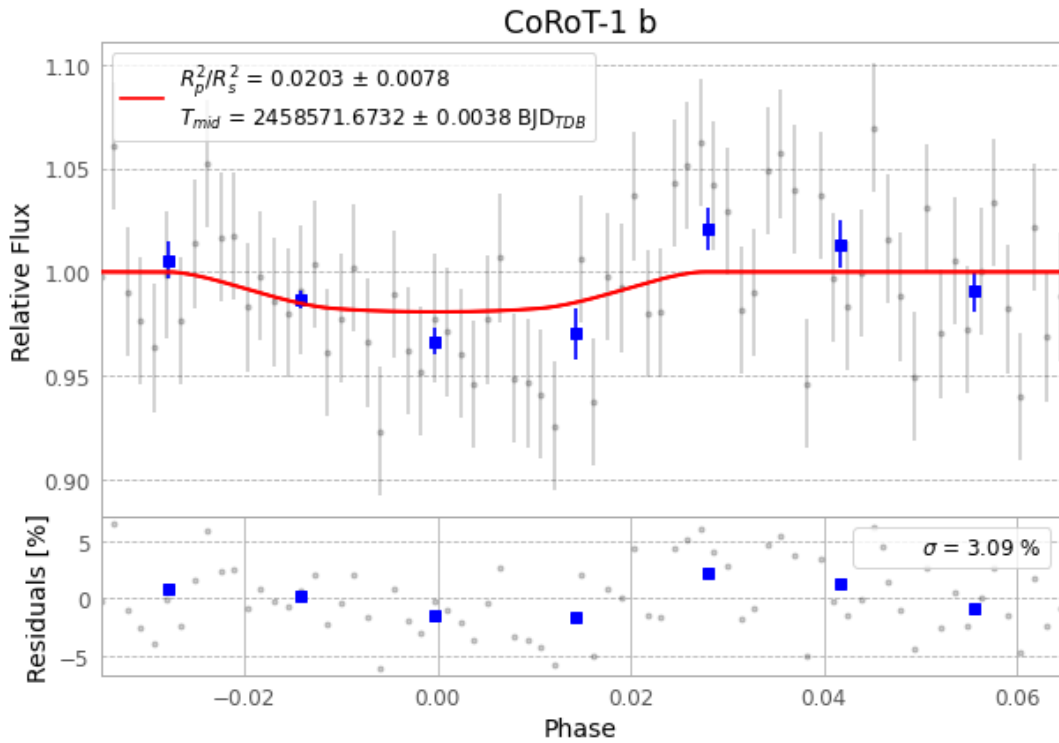


Figure 1 — FinalLightCurve_CoRoT-1 b_2019-03-28.png (SHA-256 f70b0dfd3546da1a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [322, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 db1f5685f537315f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), CoRoT-1190329025112.FITS → CoRoT-1190329062712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-190329.log (02996aa535f8...), FinalLightCurve_CoRoT-1 b_2019-03-28.png (f70b0dfd3546...), FinalLightCurve_CoRoT-1 b_2019-03-28.csv (cf9e07df36b0...)

Compute resources

Wall time 123.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)